

BANKING BACKUP SERVER

OS CONCEPT MAP • Reliable • Secure • Fault-Tolerant

OPERATING SYSTEM

Central controller of the banking backup server
Manages resources, users, processes, data and communication

↓ SUITABLE OS TYPE ↓

MULTIUSER • MULTITASKING • NETWORK / SERVER OS

Suitable for heavy concurrent backup jobs, real-time data integrity, high availability, strong security/auditing and reliable resource sharing under regulatory demands.

↓ OS SERVICES WORK TOGETHER ↓



USER MANAGEMENT

Authentication • permissions
Staff accounts • audit trail



PROCESS MANAGEMENT

Scheduling • multitasking
Backup/replication jobs • sync



MEMORY MANAGEMENT

Allocation • virtual memory
Buffering • checksum staging



FILE MANAGEMENT

Create • read • write
Encrypted archives • access control



DEVICE / I-O

Drivers • input/output
Storage arrays, tape & RAID devices



COMMUNICATION

Sockets • IPC • networking
Branch/server data exchange

↕ INTERACTION AMONG MODULES ↕

BACKUP
REQUEST

SYSTEM CALL /
KERNEL

PROCESS +
MEMORY

I/O +
NETWORK DEVICE

BACKUP
CONFIRMATION

↔ RELEVANT SYSTEM CALLS ↔

PROCESS

fork() • exec() • wait()

MEMORY

mmap() • brk() • shmget()

FILES

open() • read/write() • fsync()

NETWORK

socket() • send() • recv()

CORE GOAL → Data integrity + reliability + security under heavy backup workload